CURRICULUM VITAE – Zikai Zhang

Zikai Zhang（张子凯）

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EDUCATION

Jan. 2023-Present **Department of Computer Science and Engineering, University of Nevada, Reno, USA**
Ph.D. in Computer Science and Engineering

Sep. 2018-Jun. 2021 **Department of Computer Science and Technology, Huaqiao University, China**
M.Eng. in Computer Technology

Sep. 2014-Jul. 2018 **Department of Mechatronics and Vehicle Engineering, East China Jiaotong University, China**
B.Eng. in Mechanical Manufacturing and Automation

RESEARCH INTEREST

Distributed Machine Learning, Efficient AI, Cybersecurity, Large Language Models, Edge and Cloud Computing

TEACHING INTEREST

Artificial Intelligence, Computer Programming, Cybersecurity, Computer Vision, and Computer Networks

RESEARCH EXPERIENCE

Jan. 2023-Present **Department of Computer Science and Engineering, University of Nevada, Reno, USA**
Graduate Research and Teaching Assistant, Advisor: Dr. Rui (Zoey) Hu
- Large language model fine-tuning in distributed and resource heterogeneous environments;
- Attack and defense methods for (multimodal) large language models;
- Privacy-preserving and defense mechanisms for resilient distributed machine learning.

Sep. 2022-Nov. 2022 **Institute of Perceptual Interaction, National Virtual Reality Innovation Center, China**
Algorithm Engineer
- Efficient pupil detection algorithm for virtual-reality glasses.

Aug. 2021-Mar. 2022 **Institute of Intelligent Vehicle, Tongji University, China**
Research Assistant
- Multimodal visual understanding for autonomous systems.

Apr. 2021-Jul. 2021 **Department of Research, DMAI, Inc., Guangzhou, China**
Research Intern
- Occlusion and distortion-robust optical character recognition.

Apr. 2018-Jun. 2021 **Computer Vision and Pattern Recognition Lab, Department of Computer Science and Technology, Huaqiao University, China**
Advisor: Dr. Bineng Zhong
M.Eng Thesis: Deep Convolutional Neural Network based Long-term Object Tracking and Object Detection
- Distractor-aware, efficient, and long-term visual object tracking;
- Semi-supervised learning for object detection.

AWARD AND GRANT

Oct. 2025 **First Place Poster,** 2025 UNR Cybersecurity Conference, USA.
Sep. 2025 **Co-PI, Dynamic Language Infrastructure (DLI),** NSF, USA (Under Review).
Aug. 2025 **HDRFS Student Publication & Travel Support,** NSF EPSCoR, USA.

Oct. 2024	**First Place Poster,** 2024 UNR Cybersecurity Conference, USA.
Jan. 2024	**Co-PI, Nevada WateReuse Research Grant,** WateReuse Association**,** USA.
Oct. 2023	**2023 IEEE CNS Travel Grant,** NSF, USA.
Oct. 2023	**First Place Poster,** 2023 UNR Cybersecurity Conference, USA.
Oct. 2020	**TOP 1% (5/489)**, **Electric Vehicle Helmet Recognition Competition**, The 3rd Chinese Conference on Pattern Recognition and Computer Vision (PRCV 2020), China.
Aug. 2020	**TOP 1% (22/2245), Silver Medal**, **Kaggle Global Wheat Detection Competition**, Computer Vision Problems in Plant Phenotyping (CVPPP) Workshop, 2020 European Conference on Computer Vision (ECCV), Online.
2018-2021:	**Sole PI, Graduate Innovation Fund** in Scientific Research, Huaqiao University, China.

PUBLICATION AND PATENT

Peer Reviewed Journals and Conferences:

[1] **Zikai Zhang**, Rui Hu, Zhiyuan Yu, and Olivera Kotevska. *"STJail: Practical Video Jailbreak Attack Against Multimodal Large Language Models."* Under Review.

[2] **Zikai Zhang**, Rui Hu, and Jiahao Xu. *"SelfGrader: Detecting Jailbreak Attacks on Large Language Models with Token-Level Logit Distribution."* Under Review.

[3] **Zikai Zhang**, Rui Hu, Ping Liu, and Jiahao Xu. *"Fed-pilot: Optimizing LoRA Allocation for Efficient Federated Fine-Tuning with Heterogeneous Clients."* Under Review.

[4] **Zikai Zhang**, Rui Hu, and Jiahao Xu. *"Heterogeneous Federated Fine-Tuning with Parallel One-Rank Adaptation."* The Fourteenth International Conference on Learning Representations, ICLR (2026).

[5] Jiahao Xu, Rui Hu, Olivera Kotevska, and **Zikai Zhang**. *"Traceable Black-Box Watermarks for Federated Learning."* The Fourteenth International Conference on Learning Representations, ICLR (2026).

[6] **Zikai Zhang**, Suman Rath, Jiahao Xu, and Tingsong Xiao. *"Federated Learning for Smart Grid: A Survey on Applications and Potential Vulnerabilities."* ACM Transactions on Cyber-Physical Systems, ACM TCPS (2026).

[7] Yan Gao, Massimo Roberto Scamarcia, Javier Fernandez-Marques, Mohammad Naseri, Chong Shen Ng, Dimitris Stripelis, Zexi Li, Tao Shen, Jiamu Bai, Daoyuan Chen, **Zikai Zhang**, Rui Hu, InSeo Song, Lee KangYoon, Hong Jia, Ting Dang, Junyan Wang, Zheyuan Liu, Daniel Janes Beutel, Lingjuan Lyu, and Nicholas D Lane. *"FlowerTune: A Cross-Domain Benchmark for Federated Fine-Tuning of Large Language Models."* Advances in Neural Information Processing Systems, NeurIPS (2025).

[8] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"On the Out-of-Distribution Backdoor Attack for Federated Learning."* The 26th International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing, ACM MobiHoc (2025).

[9] **Zikai Zhang**, Ping Liu, Jiahao Xu, and Rui Hu. *"Fed-HeLLo: Efficient Federated Foundation Model Fine-Tuning with Heterogeneous LoRA Allocation."* IEEE Transactions on Neural Networks and Learning Systems, TNNLS (2025).

[10] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Detecting Backdoor Attacks in Federated Learning via Direction Alignment Inspection."* Proceedings of the Computer Vision and Pattern Recognition Conference, CVPR (2025).

[11] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Achieving Byzantine-Resilient Federated Learning via Layer-Adaptive Sparsified Model Aggregation."* IEEE/CVF Winter Conference on Applications of Computer Vision, WACV (2025).

[12] Jiahao Xu, **Zikai Zhang**, and Rui Hu. *"Backdoored Model in Federated Learning via Individual Unlearning."* IEEE/CVF Winter Conference on Applications of Computer Vision, WACV (2025).

[13] **Zikai Zhang** and Rui Hu. *"Byzantine-robust Federated Learning with Variance Reduction and Differential Privacy."* 2023 IEEE Conference on Communications and Network Security, IEEE CNS (2023).

[14] Zichen Liang, Hu Cao, Chu Yang, **Zikai Zhang**, and Guang Chen. *"Global-local Feature Aggregation for Event-based Object Detection on Eventkitti."* 2022 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems, IEEE MFI (2022).

[15] Siyuan Cheng, Binfei Chu, Bineng Zhong, **Zikai Zhang**, Xin Liu, Zhenjun Tang, and Xianxian Li. *"DRNet: Towards Fast, Accurate and Practical Dish Recognition."* Science China-Technological Sciences (2021).

[16] **Zikai Zhang**, Bineng Zhong, Shengping Zhang, Zhenjun Tang, Xin Liu, and Zhaoxiang Zhang. *"Distractor-Aware Fast Tracking via Dynamic Convolutions and MOT Philosophy."* Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR (2021).

Patents:

[1] Bineng Zhong, **Zikai Zhang**, Yaozong Zheng, Qihua Liang, and Xianxian Li. *"Wheat Head Detection Method based on Computer Vision Semi-supervised Pseudo Label Learning."* CN PATENT (CN113554627A, 2021).

[2] Bineng Zhong, **Zikai Zhang**, Yaozong Zheng, Zhenjun Tang, Xianxian Li, Xin Liu. *"Low-power Consumption Real-time Helmet Detection Method based on Computer Vision Target Detection."* CN PATENT (CN113128476A, 2021).

MENTORING AND TEACHING EXPERIENCE

Mentoring:

2025 Fall — **Mentor,** Research Project, **University of Nevada, Reno, USA**
- Students: Mia Fisher, Karam Alkherej
- Project: Implicit Prompt Injection Attack and Defense

2023-2024 — **Mentor,** Capstone Project for Undergraduates, **University of Nevada, Reno, USA**
- Students: Cody Long, Zachary Strazi, Kristian Konstantinov, Jacob Ayers
- Project: Automatic Speech Recognition Attack with Adversarial Examples

2023 Summer — **Mentor,** Research Experiences for Undergraduates (REU), **University of Nevada, Reno, USA**
- Students: Manny Ortiz (Penn State University), Zhang Lin (Colorado School of Mines)
- Project: Backdoor Attacks in Federated Learning Systems
- Supported by NSF #2150394

Teaching:

2025 Fall — **Graduate Teaching Assistant,** CS442/642 Cloud Computing, **University of Nevada, Reno, USA**
- Design course projects and tutorials on AWS-powered AI web applications;
- Lead office hours and discussion sessions to support students with programming assignments and course concepts.

2025 Spring — **Graduate Teaching Assistant,** CS302 Data Structures, **University of Nevada, Reno, USA**
- Lead office hours to support students with programming assignments and conceptual questions;
- Grade programming assignments and quizzes, and provide constructive feedback on algorithm design and code quality.

2019-2020 — **Sessional Lecturer,** Web Interaction Design, **Xiamen Institute of Technology, China**
- Introduction of JavaScript; JavaScript Objects and Arrays; Introduction of jQuery; jQuery Selector; jQuery DOM Object; jQuery Elements and Sets; jQuery Events and Animation; jQuery Function; Validation Engine and Regular Expression.

PROFESSIONAL SERVICE

TPC/PC Member:

The Annual AAAI Conference on Artificial Intelligence (AAAI) 2025, 2026

International Conference on Smart Mechatronics (ICSMech) 2024

International Conference on Intelligent Knowledge Systems and Engineering Applications (IKSEA) 2024

Journal Reviewer:

IEEE Transactions on Neural Networks and Learning Systems
IEEE Transactions on Dependable and Secure Computing
IEEE/ACM Transactions on Networking
Information Processing and Management
IEEE Open Journal of the Communications Society
China Communication
Journal of Information Security and Applications
IEEE Latin America Transactions
Cluster Computing
Engineering, Technology & Applied Science Research

Conference Reviewer:

The International Conference on Learning Representations (ICLR) 2026
International Conference on Machine Learning (ICML) 2025
Conference on Neural Information Processing Systems (NeurIPS) 2022, 2024, 2025
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025, 2026
International Conference on Computer Vision (ICCV) 2025
The European Conference on Computer Vision (ECCV) 2026
Conference on Language Modeling (COLM) 2026
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2025
International Conference on Pattern Recognition (ICPR) 2025
IEEE Consumer Communications & Networking Conference (CCNC) 2024, 2025
IEEE International Conference on Computer Communications and Networks (ICCCN) 2025